Exp 10 Interworking with router in CISCO Packet Tracer simulator.

a Design and configure a simple internetwork using router.

In this network, a router and 2 PCs are used computer are connected with router using a copper straight-through cable. After forming the network, to check network connectivity a simple PDU is transferred from PC0 to PC1

Procedure:

Step 1 (configuring Router1)

1. Select the router and open CLI

2. Press Enter to start configure Router

3. Type enable to activate the Privileged mode.

Router1 command line Interface:

Router > enable

Router # config t

Router (config-if)#

%. Link-5 changed : Interface Gigabit Ethernet 0/0 changed state to up Router (config-if)

interface Fast Ethernet 0/1

Router(config-if)# ip address 192.168.20.1

Step 2 (configuring PCs)

1. Assign Ip Address to every PC in the network

2. Select the PC. Go the desktop and select IP configuration and assign an IP address, Default gateway. Subnet Mask

3. Assign the default gate way of PC0 as 192.168.10.1

4. Assign the default gateway of PC0 as 192.168.10.1

Step-3 (connecting PC's with Router):

1. Connect Fast Ethernet 0 port of PC 0 using copper Straight Through cable Fast Ethernet 0/0 port of Router

2. Connect fast Ethernet 0 Port of PC 1 with Fast Ethernet 0/1 port of Router using copper straight-through cable

Router configure Table

Device Name	Ip address Fast Ethernet	Subnet Mask	Ip Address Fast Ethernet 0/1	Sub Mo
Router 1	192.168.10.1	255.255.255.0	192.168.20.1	255. 25

PC configuration table

Device Name	IPaddress F.	Subnet Mask	Gateway
PC0	192.168.10.2	255.255.2550	192.168.10
PC1	192.168.20.2	255.255.255.0	192.168.20

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  administratively down down
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

```
Router(config)#int gigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

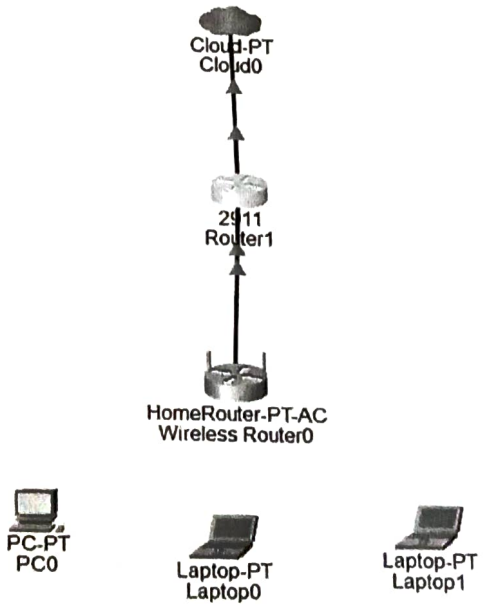
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

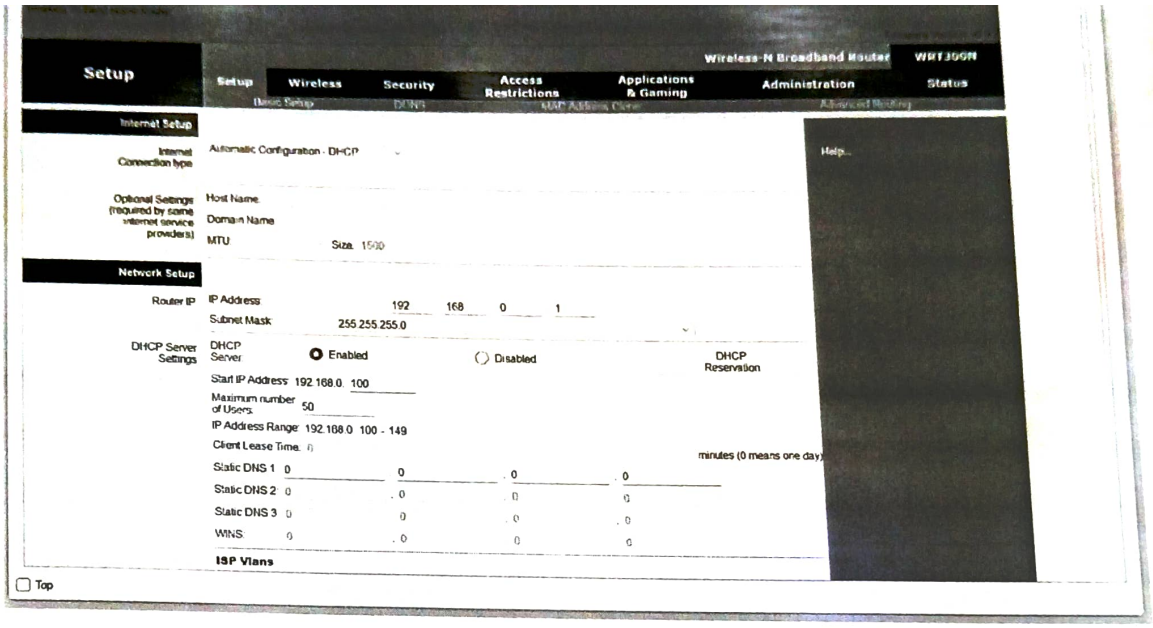
Router(config-if)#exit
Router(config)#end
```

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.1        YES manual up                    up
GigabitEthernet0/1     192.168.1.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#copy running-config sta-config
                           ^
% Invalid input detected at '^' marker.

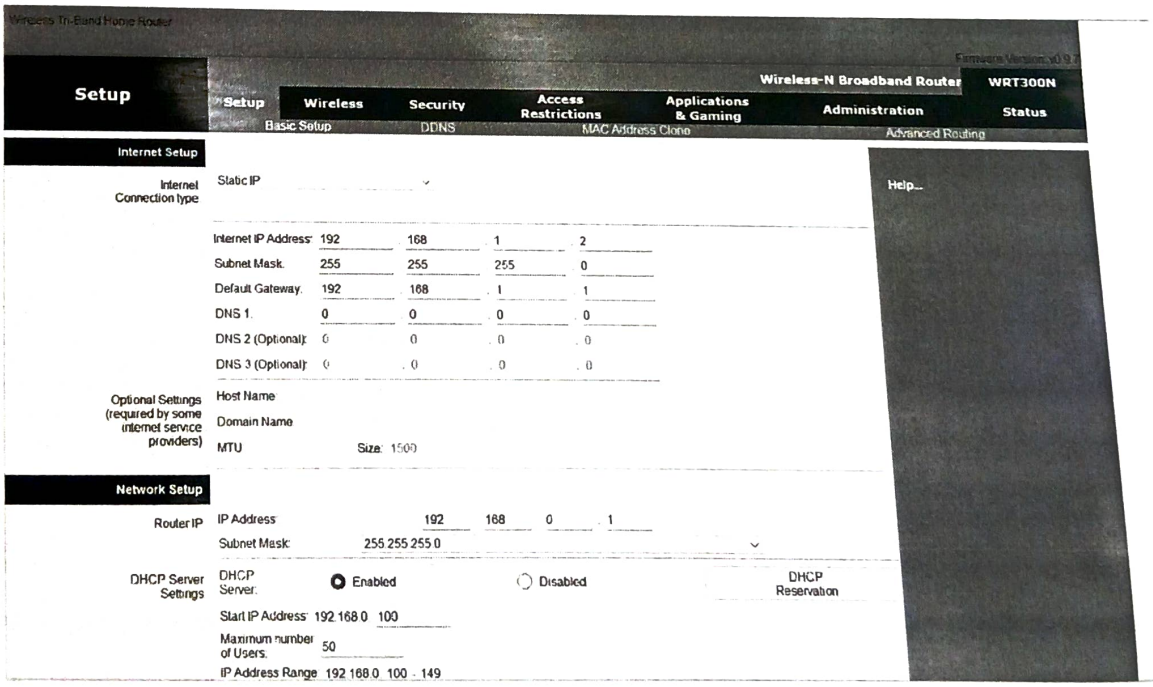
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

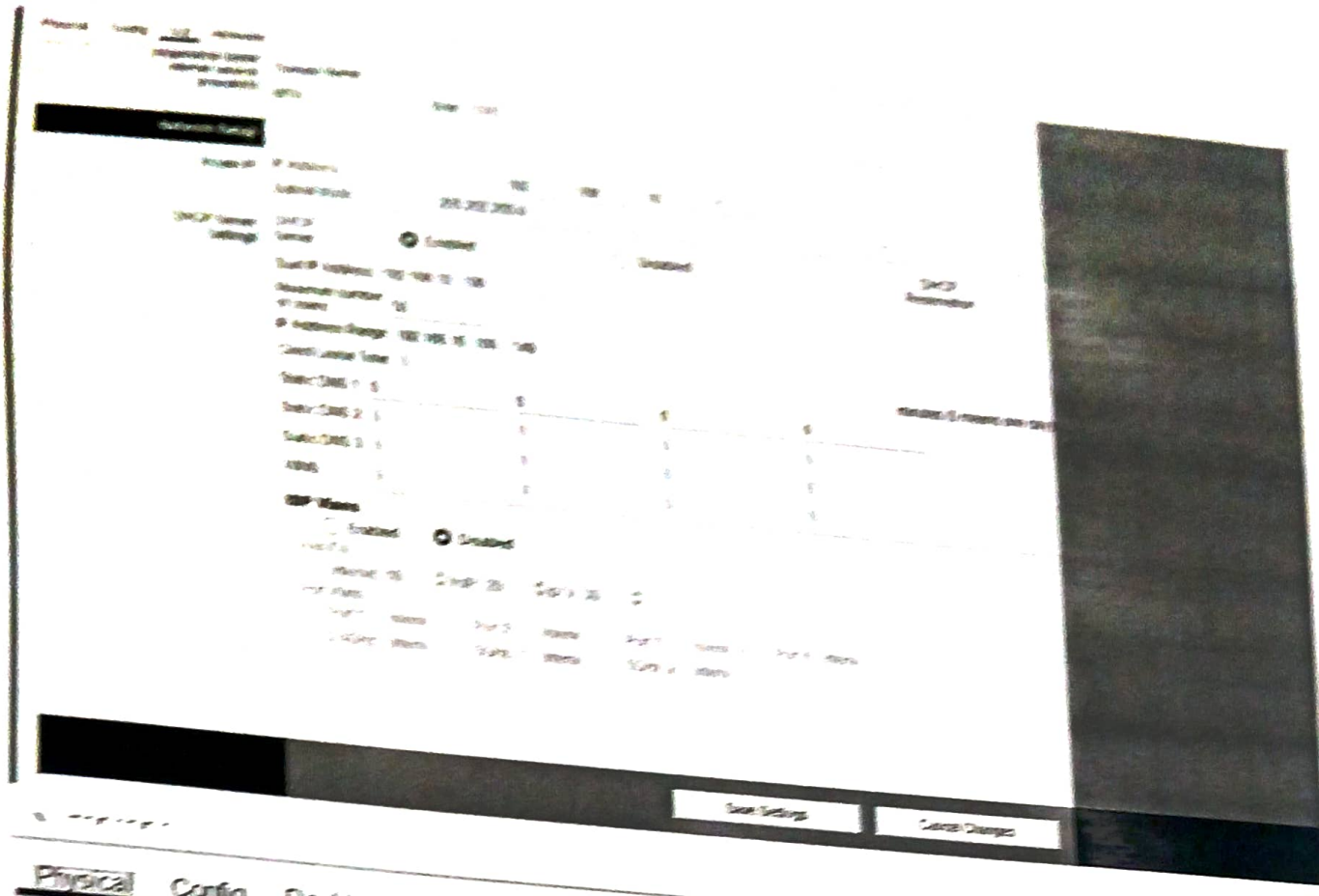
```
!
!
!
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 10.0.0.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
```





Physical Config GUI Attributes





Physical Config Desktop Programming Attributes

MODULES

WPC300N

PT-LAPTOP-NM-1CE

PT-LAPTOP-NM-1CFE

PT-LAPTOP-NM-1CGE

PT-LAPTOP-NM-1FFE

PT-LAPTOP-NM-1FFE-SM

PT-LAPTOP-NM-1FGE

PT-LAPTOP-NM-1FGE-SM

PT-LAPTOP-NM-1W

PT-LAPTOP-NM-1W-A

PT-LAPTOP-NM-1W-AC

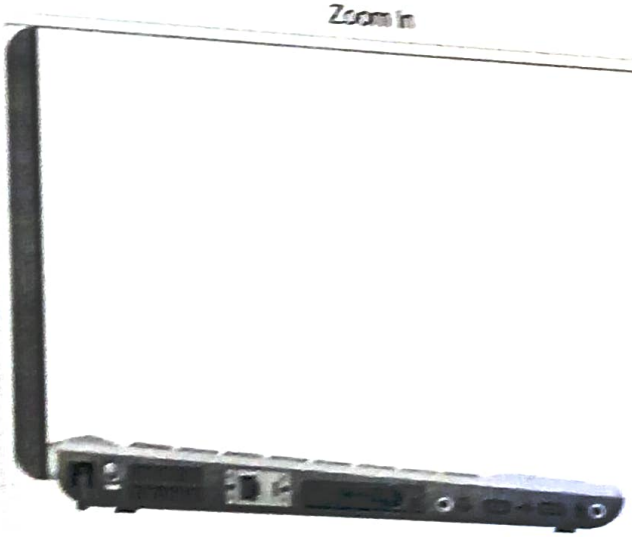
PT-LAPTOP-NM-3G/4G

PT-LAPTOP-NM-5G

PT-HEADPHONE

PT-MICROPHONE

Zoom In



Result: Thus the wireless router, DHCP server and internet cloud were successfully configured and connectivity to the cisco.com server was verified